

A3: Databases - Exam Lab



This exam practice page is designed for the A3: Databases unit of the new IB Computer Science syllabus (first assessment 2027).

Syllabus Links:

[A3.1 Database fundamentals](#)

[A3.2 Database design](#)

[A3.3 Database programming](#)

[A3.4 Alternative databases and data warehouses \(HL only\)](#)

The Exam Scenario

Scenario: "GlobalShop" E-commerce Platform. GlobalShop is a fast-growing international online retailer. Currently, they use a central **Relational Database Management System (RDBMS)** to manage their inventory, customer data, and order processing. The database contains tables such as **CUSTOMERS**, **ORDERS**, **PRODUCTS**, and **ORDER_LINE_ITEMS**. Customers have unique IDs and unique email addresses. An order can contain multiple different products in varying quantities. A product has a unique SKU (Stock Keeping Unit). To support their global growth, GlobalShop is considering moving from a centralised database to a **distributed database** architecture.

Core Questions (HL and SL)

Give these questions a try! If you're stuck, check out the 'Strategy' section to help you get started. Once you're finished, use the Mark Scheme and Worked Example to review your progress. If these feel a bit tough right now, head back to the eBook pages to sharpen your understanding first.

Question 1: Data Modeling & Integrity (SL & HL)

(a) Based on the scenario, **construct** an Entity-Relationship Diagram (ERD) showing the relationship between **ORDERS**, **PRODUCTS**, and **ORDER_LINE_ITEMS**. Specify the **cardinality** of the relationships. **[3 marks]**

(b) **Identify** appropriate **Primary Keys (PK)** for the **ORDERS** and **ORDER_LINE_ITEMS** tables. **Suggest a Composite Primary Key** if necessary and explain your choice. **[3 marks]**

Level 1: Strategy

Level 2: Mark Scheme

Level 3: Worked Example

Question 2: Normalisation (SL & HL)

GlobalShop initially used a flat file to track product information in its regional warehouses, resulting in redundant data:

Product_Name	SKU (PK)	Warehouse_Location	Warehouse_Address	Qty_in_Stock
UltraTab 10	SKU-001	London_Central	1 Main St, London	500
Case 10	SKU-A05	London_Central	1 Main St, London	1200
UltraTab 10	SKU-001	NYC_East	5 River Rd, NYC	300

(a) The table above contains **redundant data**. **Describe** one operational problem that could occur because of this redundancy if the London warehouse changes its address. **[2 marks]**

(b) **Construct** the resulting tables after **normalising** the table above to Third Normal Form (3NF). You must explicitly show the **Primary Keys (PK)** and **Foreign Keys (FK)**. **[6 marks]**

Level 1: Strategy

Level 2: Mark Scheme

Level 3: Worked Example

HL Only: Advanced Networking

Give these questions a try! If you're stuck, check out the 'Strategy' section to help you get started. Once you're finished, use the Mark Scheme and Worked Example to review your progress. If these feel a bit tough right now, head back to the eBook pages to sharpen your understanding first.

Question 3: Transactions & ACID (HL Only)

During GlobalShop's "Black Friday" sale, two different users try to purchase the exact same "limited stock" item at the same microsecond. The stock level is currently 1. User A initiates a transaction (T1), and User B initiates a transaction (T2).

- (a) **Define** the term **Database Transaction**. [1 mark]
- (b) The system must adhere to **ACID properties**. **Evaluate** why the property of **Consistency** is essential in this scenario. [3 marks]
- (c) **Explain** how a **concurrency control strategy** is used by the DBMS to ensure the property of **Isolation** between T1 and T2. [3 marks]

Level 1: Strategy

Level 2: Mark Scheme

Level 3: Worked Example

Question 4: Database Architectures (HL Only)

GlobalShop is considering upgrading its current centralised database to a **distributed database** architecture to support regional operations.

- (a) GlobalShop wants to implement **horizontal fragmentation**. **Describe** what is meant by horizontal fragmentation in this context. [2 marks]
- (b) **Discuss** the strengths and limitations of GlobalShop implementing a **fully replicated** distributed database architecture compared to their current centralised system. [6 marks]

Level 1: Strategy

Level 2: Mark Scheme

Level 3: Worked Example

How confident do you feel with this topic?

Exam Ready: I answered most questions correctly without looking at the eBook. I can explain these concepts to someone else and understand the "why" behind the mark scheme.

Move on to the next topic.

Practitioner: I understood the general concepts, but I missed a few marks on specific terminology or complex applications. I needed the "Strategy" guide for the harder questions.

Annotate your answers with the Mark Scheme and revisit the eBook sections where you lost marks. You can also use the bookmarking and notes feature on the eBook pages to highlight certain areas.

Novice: I struggled to get started on the questions. I found it difficult to apply the theory to the practice scenarios.

Go back to the eBook and Worked Examples before attempting these questions again. Perhaps use some of the Quizlets to practice the key terms and review the command terms.

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